

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457778.87121 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.097 +/- 0.045
Transit depth (Rp/Rs)^2	0.94 +/- 0.88 [%]
Orbital Inclination (inc)	86.78 +/- 2.89
Airmass coefficient 1 (a1)	0.031 +/- 0.013
Airmass coefficient 2 (a2)	1.66 +/- 0.26
Scatter in the residuals of the lightcurve fit is	34.92 %
Best Comparison Star	None
Optimal Aperture	3.7
Optimal Annulus	13.05
Transit Duration (day)	0.0405 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.097 ± 0.045 vs 0.1178 (deviation 0.3σ)
Duration fit vs published	0.0405 d vs 0.125 d (deviation 18.43σ)
Mid-time O–C	0.6 min
Depth SNR	1.4
Residual scatter	34.92 %

Light curve

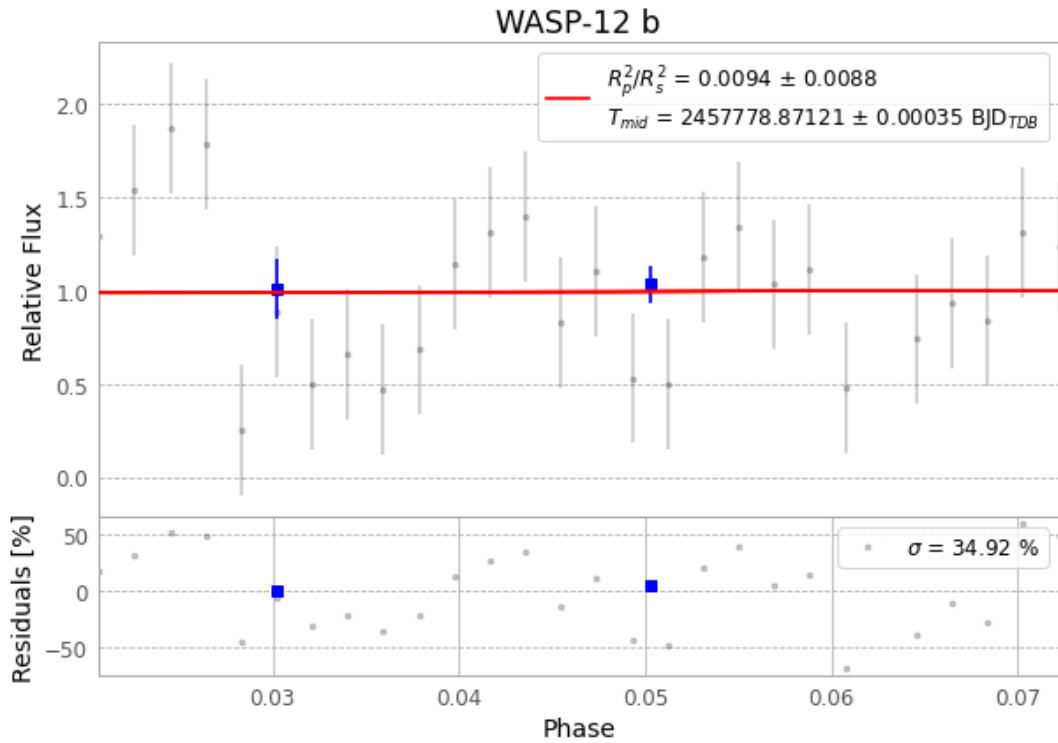


Figure 1 — FinalLightCurve_WASP-12 b_2017-01-25.png (SHA-256 fc27496e5694f0a8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [626, 414], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7ca3ab64658fa6a6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

53 FITS frames (35 MB), WASP-12170125091813.FITS → WASP-12170125115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-170125.log (9899bc261d19...), FinalLightCurve_WASP-12 b_2017-01-25.png (fc27496e5694...), FinalLightCurve_WASP-12 b_2017-01-25.csv (b9b016681ded...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 19:44Z.